

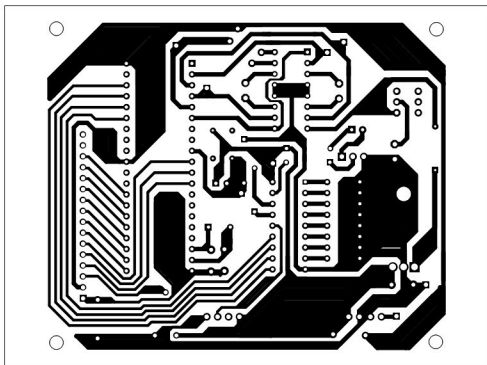


Fig. 5: An actual-size, single-side PCB for gesture transmitter

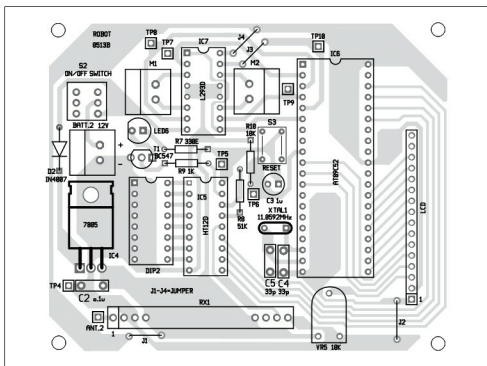


Fig. 6: Component layout for the PCB in Fig. 5

driver L293D (IC7) and a few discrete components.

Microcontroller AT89C52 is the heart of the circuit. It is a low-power, high-performance CMOS eight-bit microcontroller that has 8 kB of Flash, 256 bytes of RAM, 32 input/output (I/O) lines, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

In addition, AT89C52 is designed with static logic for operation down to zero frequency and supports two software-selectable power saving modes. Idle mode stops the CPU while allowing the RAM, timers/counters, serial port and interrupt system to continue functioning. Power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next hardware reset.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

Data transmitted from the gesture transmitter is received by RF receiver module RX1 and further decoded by decoder IC5. Make sure that identical addresses are selected for the encoder and the decoder through switches DIP1 and DIP2.

Data reception is indicated by LED6. Decoded data is fed to port pins P0.1 through P0.4 of microcontroller IC6. The outputs from port pins P1.1 through P1.4 of the microcontroller are fed to IN1 through IN4 of motor driver IC7 to drive motors M1 and M2. Outputs OUT1 and OUT2 drive motor M1 and outputs OUT3 and OUT4 drive motor M2.

Enable pins EN1 (pin 1) and EN2 (pin 9) are joined together and connected to port pin P1.5 of microcontroller IC6.

Data pins of the LCD are connected to port pins P2.0 through P2.7, and control pins RS, $R/\bar{W}$ and EN are connected to P3.2, P3.1 and P3.0, respectively. When you make a gesture to move the robot in forward direction, the LCD will show 'receiving signal' in the first line and 'going forward' in the second line. Similarly, messages are displayed for other directions.

'Power-on' reset is provided by the combination of resistor R10 and capacitor C3. Switch S3 is used for manual reset. An 11.0592MHz crystal along with two 33pF capacitors provides basic clock frequency to microcontroller IC6. Regulator IC4 is used to provide regulated 5V supply to the circuit. The 12V battery supply is directly connected to pin 8 of motor driver IC7 for V_{DD} .

Construction and testing

Actual-size, single-side PCBs for the gesture transmitter and the wireless robot are shown in Figs 5 and 7 and their component layouts in Figs 6 and 8, respectively. Assemble the circuits on PCBs as this saves time and